

I, Samii Shabuse, have read this report and attest that the contents are true and correct.
I, Mustafa Bookwala, have read this report and attest that the contents are true and correct.

I updated the Python script so that it now is a self-contained builder that constructs the entire database in one go to allow for easier project set up. I was also the lead for maintaining progress and documentation of our work through GitHub, so I ensured that all branches, merge requests, and versions were maintained. For the final part of the application, I coded the pages for Find a Ride, Available Rides, Offer a Ride, and Available Ride Offers. Samii assisted me with the database connection to these pages, and I also gave finishing touches to the presentation working with Mustafa. My part of the coding was done to support our application on Android devices.

Builder python script:

```
generate_database.py X
Find-My-Ride > database > generate_database.py > ...
1  #!/usr/bin/env python3
2  """
3  generate_database.py
4  1. Generates database/schema.sql and database/populate.sql.
5  2. Creates 'findmyride.db' from scratch in the database folder.
6  3. Executes the schema and population scripts against the database.
7  4. Copies the finished database to ./app/composeApp/src/androidMain/assets/findmyride.db so that it can be used by the application
8  """
9
10 import os
11 import random
12 import datetime
13 import sqlite3
14 import shutil
15 from pathlib import Path
16
17 random.seed(42)
18
19 # --- CONFIGURATION ---
20
21 # SQL Output Paths
22 OUT_DIR = Path("database")
23 OUT_DIR.mkdir(parents=True, exist_ok=True)
24 SCHEMA_PATH = OUT_DIR / "schema.sql"
25 POPULATE_PATH = OUT_DIR / "populate.sql"
26
27 # DB Path
28 DB_FILENAME = OUT_DIR / "findmyride.db"
29 CURRENT_DB_PATH = Path(DB_FILENAME)
30
31 # Target Asset Path
32 ASSET_DIR = Path("./app/composeApp/src/androidMain/assets")
33 ASSET_DB_PATH = ASSET_DIR / "findmyride.db"
34
35 # --- SCHEMA DEFINITION ---
36 SCHEMA_SQL = """
37 PRAGMA foreign_keys = ON;
38
39 CREATE TABLE "USER" (
40     user_id      INTEGER PRIMARY KEY,
41     email        TEXT     NOT NULL UNIQUE,
42     username     TEXT     NOT NULL UNIQUE,
43     password_hash TEXT     NOT NULL,
44     phone_number TEXT,
45     role         TEXT     NOT NULL CHECK (role IN ('rider','driver','both')),
46     rating_avg   REAL     DEFAULT 2.5 CHECK (rating_avg BETWEEN 0 AND 5),
47     created_at   TEXT     NOT NULL DEFAULT (datetime('now'))
48 )
49 """
```

Commits showing development progress:



kennanLu

31 commits 4,075 ++ 1,295 --

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